

RESEARCH INTERESTS

- Computation theory (automata, formal grammar, Chomsky hierarchy, Parikh matrix, etc.)
- Computational linguistics & NLP (grammar-based NLP, few-shot learning, data augmentation)

RESEARCH STATEMENT

- **Overview:** I am actively conducting research that integrates computation theory and natural language processing (NLP) under the guidance of my advisor, Professor Yo-Sub Han. I believe that the theory of computation extends beyond theoretical boundaries and can be applied to deep learning and NLP. I pursue interdisciplinary research that bridges theory and practice.
- **Research Interests:** My research focuses on computational linguistics, particularly utilizing Parikh matrices, automata, and grammar-based learning techniques. I am particularly interested in merging the formal languages with language models and exploring various grammatical applications in deep learning-based NLP.

EDUCATION

Yonsei University

Integrated Ph.D. course in Computer Science

Seoul, KR

2019–Current

- Research focus: computational linguistics, Parikh matrix, formal grammar-based NLP

Yonsei University

B.Eng. in Computer Science

Seoul, KR

2015–2019

EXPERIENCE

Theory of Computation Lab, Yonsei University

Research Intern

Seoul, KR

Summer, 2017–Winter, 2018

- Studied automata and formal grammars for language classification tasks
- Explored computational linguistics approaches for simplified NLP tasks

Other Research/Work Experience

2019–Current

- Bridging formal language theory and deep learning for advanced NLP
- Developing grammar-based data augmentation for few-shot NLP tasks
- Investigations on Parikh matrix, M-equivalence, and other language-theoretic properties

PROJECTS

2019–2020: EEG Rule Extraction & Data Similarity Approximation

- Wrote a proposal for the project on EEG stimulation data rule extraction upon entering graduate school in 2019, and led the project.
- Wrote proposals and presentation materials for StarLab funding (2020–2022) on using approximate algorithms for data similarity measurement.

Malware Pattern Extraction & Human-AI Programming Platform Research Project

- Conducted machine learning-based malware pattern extraction and participated in the Human-AI Programming Platform project.
- Researched synergies between NLP and source code, such as code summarization and complexity analysis.

2022–2022: 5th AI Grand Challenge

- Developed a deep learning model for solving mathematical equation problems in a three-stage national-level competition.
- Passed the first stage and wrote the proposal for the competition project.
- Participated in the second-stage competition.

2023–2024: AI Model Learning Trend Analysis for Financial Data

- Collaborated with Bankware Global, leading and planning the research project.
- Analyzed AI models trained on financial data and explored new research directions.

PUBLICATIONS

(To appear)

Characterizations of M-equivalence and weak M-relation

- Joonghyuk Hahn[†], Hyunjoon Cheon[†], Yo-Sub Han
- *To appear in International Journal of Foundations of Computer Science (IJFCS)*
- **(co-first author)**
- Formalizes M-equivalence and weak M-relation over Parikh matrices, extending theoretical foundation for language classification.

TCPProF: Time-Complexity Prediction SSL Framework

- Joonghyuk Hahn[†], Hyeseon Ahn, Jungin Kim, Soohan Lim, Yo-Sub Han
- *To appear in NAACL'25*
- **(first author)**
- Proposes a novel semi-supervised learning framework for code time complexity prediction, integrating symbolic analysis in low-resource settings.

2024

On the Decidability of Infix Inclusion Problem

- Hyunjoon Cheon, Joonghyuk Hahn[†], Yo-Sub Han
- *Theory of Computing Systems, Vol. 68, 301–321 (2024)*
- **(co-first author)**
- Investigates theoretical properties of infix inclusion problems, establishing new decidability results in formal language theory.

Universal Rewriting Rules for the Parikh Matrix Injectivity Problem

- Ingyu Baek, Joonghyuk Hahn[†], Yo-Sub Han, Kai Salomaa
- *Proceedings of DLT'24, LNCS 14791, 68–81*
- **(co-first author)**

- Derives universal rewriting rules to tackle the injectivity question in Parikh matrix frameworks, advancing theoretical results in language rewriting.

SuperST: Superficial Self-Training for Few-Shot Text Classification

- Ju-Hyoung Lee, Joonghyuk Hahn[†], Hyeon-Tae Seo, Jiho Park, Yo-Sub Han
- *Proceedings of LREC-COLING 2024*, 15436–15447
- **(co-first author)**
- Introduces a selective self-training method, leveraging partial learning on noisy data while fully tuning on reliable data for improved few-shot classification.

2023

ATHENA: Mathematical Reasoning with Thought Expansion

- J.B. Kim, Hazel Kim, Joonghyuk Hahn, Yo-Sub Han
- *Proceedings of EMNLP 2023*, 16315–16327
- **(contributing author)**
- Enhances mathematical reasoning via a structured “thought expansion” approach in large language models.

GDA: Grammar-based Data Augmentation for Text Classification using Slot Information

- Joonghyuk Hahn[†], Hyunjoon Cheon, Elizabeth G. Orwig, Su-Hyeon Kim, Sang-Ki Ko, Yo-Sub Han
- *Findings of EMNLP 2023*, 7291–7306
- **(co-first author)**
- Presents a grammar-based augmentation that leverages slot annotations to generate semantically consistent and diverse training data.

M-equivalence of Parikh Matrix over a Ternary Alphabet

- Joonghyuk Hahn[†], Hyunjoon Cheon, Yo-Sub Han
- *Proceedings of CIAA’23, LNCS 14151*, 141–152
- **(co-first author)**
- Examines M-equivalence criteria under ternary alphabets, expanding theoretical understanding of language-class equivalences.

2022

Boosting Code Summarization by Embedding Code Structures

- Jikyeong Son, Joonghyuk Hahn[†], Hyeon-Tae Seo, Yo-Sub Han
- *Proceedings of COLING’22*, 5966–5977
- **(co-first author)**
- Incorporates program dependency graph embeddings to improve code summarization performance, capturing structural semantics.

On the Decidability of Infix Inclusion Problem

- Hyunjoon Cheon, Joonghyuk Hahn[†], Yo-Sub Han
- *Proceedings of DLT’22, LNCS 13257*, 115–126
- **(co-first author)**
- Analyzes infix inclusion within formal language theory, presenting decidability results and proof techniques.

2021

Self-Training using Rules of Grammar for Few-Shot NLU

- Joonghyuk Hahn[†], Hyunjoon Cheon, Kyuyeol Han, Cheongjae Lee, Junseok Kim, Yo-Sub Han
- *Findings of EMNLP 2021*, 4576–4581
- (co-first author)
- Employs grammar rules as a reliable mechanism for pseudo-labeling in self-training, boosting performance in few-shot NLU tasks.

Most Pseudo-copy Languages Are Not Context-free

- Hyunjoon Cheon, Joonghyuk Hahn[†], Yo-Sub Han, Sang-Ki Ko
- *Proceedings of COCOON'21*, 189–200
- (co-first author)
- Investigates pseudo-copy languages, proving that most of them lie outside the context-free class via formal language analysis.

TEACHING / MENTORING

Teaching Assistant: Automata and Formal Languages

Yonsei University

2019–2022

- Led discussion sessions and grading assignments in advanced automata theory course.

Mentoring Undergraduate Research Interns

Theory of Computation Lab

2021–2023

- Guided undergraduates on formal grammar concepts and mini-projects in computational linguistics.

PROFESSIONAL ACTIVITIES

Reviewer/PC Member

2022–Current

- Reviewed manuscripts for conferences: e.g. ACL Rolling Review, EMNLP, etc.

SKILLS

- **Programming:** C/C++, Python, Java
- **Tools/Techs:** L^AT_EX, Git, PyTorch, TensorFlow, etc.

REFERENCES

Prof. Yo-Sub Han

Yonsei University

Email: emmous@yonsei.ac.kr

- Advisor, Department of Computer Science